

Algorithm arena

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.1 Algorithms — Paper 2, Computational thinking, algorithms and programming
Lesson	Algorithm arena (lesson 13 of the unit)
Lesson ID	al - bonus
Experience ID	none — this lesson is on paper
Plan ID	PLAN - AL - BONUS
Content version	3d94b613074d
Website	https://revise360.co.uk/experience.html?id=al-bonus

Big question, outcomes and success criteria

this lesson states no outcomes of its own, which is correct for what it is: a bonus challenge. It stretches outcomes already taught and is optional.

Prerequisites

The lesson before this one is al - l12 - test. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: al - l01, al - l02, al - l03, al - l04, al - l05, al - l06, al - l07, al - l08, al - l09, al - l10, al - l11.

Assessable outcomes

this lesson states no outcomes of its own, which is correct for what it is: a bonus challenge. It stretches outcomes already taught and is optional.

Vocabulary, resources and preparation

No new vocabulary: this lesson introduces nothing new. The vocabulary it draws on is listed on the earlier lessons' plans and in the unit Big Picture.

Resources

ONLINE EXPERIENCE [experience.html?id=al-bonus](https://revise360.co.uk/experience.html?id=al-bonus)

Before the lesson

1. Print the paper, one per pupil, and the reflection sheet to hand out afterwards.
2. Have the answer sheet to hand. It is teacher-only and is in the unit teacher guide pack, not in this plan.
3. Decide whether you are marking in the lesson or collecting the papers in.

Nothing online is needed. Pupils do not sign in for this lesson, and there is no experience to launch.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Set the challenge	Say what finishing it looks like.	Listen, then start.	Everyone knows what finished looks like.
5-50	The challenge	Pupils work on it; you circulate.	Work on the challenge.	Something on screen or on paper by halfway.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
50-60	Share and review	Two or three pupils show what they did and what they changed on the way.	Two or three explain what they changed.	A decision described, not just a result.

The explanation and the worked example

The wall panels of this experience are painted into its 360 image and are not held as text, so this plan cannot quote them. Open the experience yourself before teaching it: the station names are below, and the slides carry the same teaching.

final. Algorithm arena

Retrieval and the check for understanding

This is a bonus challenge, not a taught lesson. It has no starter, no new vocabulary and no hinge question because it teaches nothing new: it stretches what the unit has already taught, and it is optional. Use it with a class that has finished, or as something to come back to.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
	Every activity in this lesson is recognition: a pupil chooses between options and never produces anything of their own. Plan the independent work yourself - the exam practice on the worksheet is the obvious place - and do not read a full score here as evidence that a pupil can do it unaided.	

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

This lesson has no online experience: it is a bonus challenge. Everything it needs is on paper.

Assessment, evidence and unfinished work

Nothing on this lesson is marked. It is a challenge: the evidence is what the pupil produced and what they can say about the decisions they made.

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/al-bonus.json`, which is not published with the website.

Common errors

None are recorded for this lesson beyond the misconceptions listed above.

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

Ask the pupil to say what they changed and why. A challenge is evidenced by the decision, not the result.

Retrieval link to next lesson

This is the last lesson of the unit. The next retrieval of it is the unit test.

After this lesson

Nothing is set from a challenge. It is optional, and a pupil who did not reach it has not fallen behind.

Shorter lessons, longer lessons and no internet

A shorter lesson

This is optional work. Use it when there is time, and stop when the lesson stops; nothing is lost by leaving it unfinished.

If the internet or the devices are unavailable

Set the challenge on paper from the slides, or leave it for another lesson. There is no offline version of the experience and this plan does not contain one.

Revise 360 — a product of Revise 360 Ltd, created by Olly Pettitt. An independent resource: not written, endorsed or approved by OCR or any exam board. Classroom use and personal study only.